

Solving Process for the Complete BMPV Four-Derivative Correction

Version 5: a chronological human-readable derivation

GPT 5.6 Sol

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Abstract

This document records how the generic unequal-charge BMPV correction was derived, rather than only displaying the answer. It follows the BMPV calculation from construction of thirteen explicit radial equations, Page and constraint reduction, the equal-charge benchmark, generic rational reconstruction at orders j, j^2, j^3 , and exact closure of the rotation hierarchy. It also reconstructs the earlier static solution in its harmonic/isotropic-like paper radius and proves the exact radial map to the areal gauge used by the solver. Every stage is labelled as a discovery device, an archival equation source, or a final symbolic proof. A new literal from-EOM notebook recomputes the decisive coefficients instead of trusting recorded zero ledgers. The companion report contains the detailed physical formulas and the subsequent exact generic-static, Page/radial, Wald–Tachikawa, horizon, Cassani/heterotic, and Eq. (3.31) certificates. The direct entropy calculation precedes the comparison of published conventions. The later ten-to-five-dimensional discussion follows the field transformations in their actual order. Its scalar-ratio, winding-vector, and Lorentz–Chern–Simons sectors are established, and the generic static scalar match is symbolic. Five rotating scalar samples pass exactly in r . The later generic certificate upgrades these samples to an arbitrary-charge symbolic proof. The historical three-form-first third-vector candidate is nonclosed, but a raw-potential-first calculation proves the physical duality identity independently of the target and gives a closed solution-space map whose final potential residual vanishes symbolically for arbitrary unequal charges and rotation. The map is exact on this ansatz; a unique universal off-shell covariant derivation of its two coefficients is not claimed. Finally, a retrospective parity-odd audit distinguishes v1’s ambiguous prose from its actual determinant factors and finds that no computed five-dimensional sign must be reversed.

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1 What was being solved

For every field in the cohomogeneity-one STU/BMPV ansatz,

$$\mathcal{Y}(r, j) = \mathcal{Y}_0(r, j) + \lambda_3 \delta \mathcal{Y}(r, j) + \mathcal{O}(\lambda_3^2), \quad (1)$$

where $\mathcal{Y}$ denotes any ansatz field and $\mathcal{Y}_0$ its two-derivative rotating three-charge background. The active four-derivative direction is one complete supersymmetric standard-Weyl curvature-squared invariant, denoted λ_3 . Its vector and mixed Chern–Simons completion terms are retained.

After the allowed field redefinitions, the Cassani-frame description of the rotating BPS problem depends on this same single coupling. The heterotic normalization used later is $\lambda_1 = \lambda_2 = 0$ and $\lambda_3 = \alpha'/8$. This coupling identification is separate from the field-by-field ten-to-five-dimensional map discussed only after the five-dimensional solution and entropy are complete.

The practical target was not initially assumed to be exact in rotation. We began with the parity hierarchy

$$\delta \mathcal{Y}_{\text{even}}(r, j) = \delta \mathcal{Y}^{(0)}(r) + j^2 \delta \mathcal{Y}^{(2)}(r) + j^4 \delta \mathcal{Y}^{(4)}(r) + \cdots, \quad (2)$$

$$\delta \mathcal{Y}_{\text{odd}}(r, j) = j \delta \mathcal{Y}^{(1)}(r) + j^3 \delta \mathcal{Y}^{(3)}(r) + j^5 \delta \mathcal{Y}^{(5)}(r) + \cdots, \quad (3)$$

then let the equations decide where it terminated. The final result is first order in λ_3 , exact in j , with $\delta \mathcal{Y}^{(2n)} = 0$ for $n \geq 2$ and $\delta \mathcal{Y}^{(2n+1)} = 0$ for $n \geq 2$.

1.1 Human notation used in this document

The physical charge parameters are μ_I , while the conserved charges are $Q_I = \pi \mu_I / (4G_5)$. The background harmonic functions are

$$H_I(r) = 1 + \frac{\mu_I}{r^2}. \quad (4)$$

The three constrained scalars are represented by the two independent fields used in the original STU paper,

$$X^1 = e^{-\varphi_1/\sqrt{6}-\varphi_2/\sqrt{2}}, \quad X^2 = e^{-\varphi_1/\sqrt{6}+\varphi_2/\sqrt{2}}, \quad X^3 = e^{2\varphi_1/\sqrt{6}}, \quad X^1 X^2 X^3 = 1. \quad (5)$$

Latin indices denote the oriented orthonormal frame. The mostly-plus flat-symbol convention is

$$\epsilon_{01234} = +1, \quad \epsilon^{01234} = -1. \quad (6)$$

The curved Levi–Civita tensor instead contains the vielbeins,

$$\epsilon_{\mu\nu\rho\sigma\lambda} = e_\mu{}^a e_\nu{}^b e_\rho{}^c e_\sigma{}^d e_\lambda{}^e \epsilon_{abcde}, \quad e = \sqrt{-g}. \quad (7)$$

The BMPV base orientation is

$$\text{vol}_4 = \frac{r^3}{8} \sin \theta \, dr \wedge d\theta \wedge d\psi \wedge d\phi = -\frac{r^3}{8} dr \wedge \sigma_L^1 \wedge \sigma_L^2 \wedge \sigma_L^3. \quad (8)$$

Consequently the notebook coordinate order $(t, r, \theta, \phi, \psi)$ is orientation reversing and

$$\epsilon_{tr\theta\phi\psi} = -\sqrt{-g}, \quad \epsilon^{tr\theta\phi\psi} = +\frac{1}{\sqrt{-g}}. \quad (9)$$

Mathematica’s raw array is the alternating coordinate density

$$\tilde{\epsilon}^{tr\theta\phi\psi} = +1, \quad \tilde{\epsilon}^{\mu\nu\rho\sigma\lambda} = e \, \epsilon^{\mu\nu\rho\sigma\lambda} \quad (10)$$

in this chart. Notebook 00A constructs the orientation-reversing vielbein, proves both the frame and coordinate relations, and checks $d \star F + F \wedge F = 0$ explicitly. The archived positive coordinate density is therefore correct, and no gauge-potential sign is changed.

Version 1 did not state this symbol–tensor–density distinction clearly in its prose, so the v4 decision was made from the sealed code. Its action densities use the raw alternating array, while curved contractions use that array divided by e . In particular, the complete $W'_{IJKL} \epsilon F^I F^J F^K dX^L$ sector has the seven source coefficients and overall plus sign transcribed correctly. The mixed-curvature surface term uses the array divided by $2e$. Notebook 29 reproduces these relations and the positive Eq. 4.1 vector sign in a fresh kernel. The verdict is documentary ambiguity, not a parity-odd sign error; no v1 physical result is reversed.

For orientation, the two-derivative rotating gauge potentials are aligned with

$$\Theta_0 = dt + \frac{j}{2r^2} \sigma_L^3, \quad A_0^I = H_I^{-1} \Theta_0. \quad (11)$$

Write the corrected ansatz as

$$A^I = (H_I^{-1} + \lambda_3 \delta u_I) \left[dt + \left(\frac{j}{2r^2} + \lambda_3 \delta \omega \right) \sigma_L^3 \right] + \lambda_3 \beta_I \sigma_L^3 + \mathcal{O}(\lambda_3^2). \quad (12)$$

Only now is β_I defined: it is the genuine Hopf-fibre gauge correction after subtracting the angular terms induced by δu_I and $\delta \omega$. If one instead writes $A^I = u_I dt + v_I \sigma_L^3$, then

$$\beta_I = \delta v_I - \frac{j}{2r^2} \delta u_I - H_I^{-1} \delta \omega. \quad (13)$$

The report uses β_I because it exposes rotation parity and is not confused with the ten-dimensional Kalb–Ramond two-form. Internal notebook symbols are translated only in the final crosswalk.

1.2 The two radial coordinates

Before following the calculation, distinguish the paper radius from the solver radius. The generic static solution of arXiv:2502.05065 is written in a harmonic coordinate ρ inherited from its multi-centre construction. After the field redefinitions used there, the spatial metric has a distinguished radial correction and is therefore isotropic-like rather than strictly isotropic. The rotating calculation instead fixes the angular-area correction to zero and uses an areal/Schwarzschild-like radius r .

At first order these descriptions are related by

$$\rho = r + \lambda_3 \xi(r), \quad \xi(r) = -\frac{\delta \mathcal{H}_{\text{paper}}(r)}{\mathcal{H}'_0(r) + 1/r}, \quad \mathcal{H}_0 = \frac{1}{6} \log(H_1 H_2 H_3). \quad (14)$$

For a scalar-like ansatz function Y , $\delta Y_{\text{areal}} = \delta Y_{\text{paper}} + \xi Y'_0$. The radial metric has the additional Jacobian term $+\xi'$, and the angular correction obeys

$$\delta \mathcal{H}_{\text{areal}} = \delta \mathcal{H}_{\text{paper}} + \xi(\mathcal{H}'_0 + 1/r) = 0. \quad (15)$$

Notebook 28 checks the published α' coefficients, proves this map and its inverse for every generic unequal-charge static field, and derives the general scalar, covector and metric pullbacks without expanding in j . Applying the same j -independent inverse map to the complete rotating answer gives a valid paper-matched rotating coordinate, but not a uniquely defined fully isotropic rotating gauge. This radial diffeomorphism contains no epsilon tensor and is independent of the parity-odd audit.

2 BMPV workflow and notebook map

The notebooks supplied with these supplementary materials form a definite dependency chain. They implement stages of one BMPV/STU calculation rather than separate Lagrangians. Their common logic is

ansatz and action $\longrightarrow$ Euler–Lagrange equations $\longrightarrow$ static background insertion $\longrightarrow$ linearized correction equations $\longrightarrow$ staged elimination and integration.

This workflow is applied directly to the rotating ansatz. The Gutowski–Reall family and its ungauged STU/BMPV limit supply the background conventions; the published static four-derivative solution supplies $j = 0$ regression data. No external notebook collection is needed to reproduce the calculation: the action, background, ansatz, boundary conditions, physical ensemble, equations, and decisive certificates are all stated or included in these supplementary materials.

The notebook map in these supplementary materials is:

stage	notebook	role and evidence
00A	Levi–Civita convention	flat symbol, curved tensor, alternating density, BMPV orientation, positive vector sign, and Page/Tachikawa signs
29	retrospective v1 parity audit	sealed-v1 AFF , $W'\epsilon FFF dX$, and ARR object types, signs, determinant factors, and the no-sign-reversal verdict
00	Ansatz, action and EOM derivation	geometry, invariant tables and linearization scaffold; full curvature/action variation is resource-bounded
01	Explicit 13 EOM with sources	literal homogeneous operators and background source terms

stage	notebook	role and evidence
02	Page and radial reduction	electric first integrals, adapted variables, areal gauge and algebraic radial constraint
03	Source sectors	interpretation of the standalone vector invariant and the complete supersymmetric source
04–05	Equal-charge solution and 13/13 check	exact benchmark before generic charge reconstruction
06–07	Generic j solution and normalization	rotation-odd subsystem, fixed physical charges, and asymptotic rotation-mode normalization
08	Generic j^2 system	reduced system and all eight even fields
09	Generic j^3 system	four odd equations and closed solution
10	closure	source-degree exhaustion and j^4 – j^8 zeros
11	combined solution	all fields and the final validation ledger
12	true from-EOM verifier	literal low-order equations and 35 separately classified higher-order coefficient pairs
13	generic static certificate	exact $j = 0$ 13/13 residuals and the embedded coordinate map
28	paper-radius coordinate certificate	seven published-coefficient checks, exact generic static map in both directions, and general exact-in- j tensor pullbacks
14	generic Wald–Tachikawa certificate	complete angular surface charge, Komar calibration, and exact asymptotic tails
15	horizon certificate	common shift, ingoing co-rotating metric, finite scalar/gauge invariants, and extremality
16	Cassani/heterotic ledger	independent single- λ_3 invariant, rotating BPS source cancellation, raw dictionary and Eq. (3.31) ODEs
17	general radial certificate	full- j Page-reduced identity, physical-domain coefficient, and direct unreduced j^2 residual
20	direct five-dimensional entropy	corrected area, horizon-adapted binormal, covariant Wald kernel, and mixed Chern–Simons term
21	independent entropy reconstruction	coordinate Riemann tensor, normal connection, Lorentz-trace sign, and a six-sector pass ledger
18	entropy-convention comparison	exact algebra relating four published spin factors and the direct heterotic expansion
19	historical rotating heterotic-to-STU candidate	exact raw KK reduction, scalar-ratio cancellation, self-dual correction, and the initial evidence-aware ledger
22	corrected rotating winding/Lorentz-CS certificate	mixed-index CS, source torsion sign, all-ten vertical components, mapped KK shift, closure, and zero-tail potential
23	complete ten-component corrected-dualization source	four sectors evaluated before summation, componentwise recombination, normalization, and regression against the incomplete monolithic output

stage	notebook	role and evidence
24	field-map convention and static anchor	horizontal sign and source ledger, all ten components of the predual correction three-form, the two field-redefinition directions, and unchanged α' sign
25	independent lower-two-form certificate	exact local representative, all-ten derivative, closure, and forward-versus-inverse static sign
26	corrected postdual scalar map	generic static symbolic identity and five rotating exact-in- r parameter certificates
27	rejected three-form-first route	exact nonclosure residual and rank exclusions for the tested tied-profile repair
30	generic rotating scalar proof	arbitrary-charge symbolic cancellation of the corrected scalar residual, exact in rotation
31	raw-first rotating third-vector target match	target-independent raw duality for arbitrary radial profiles, physical orientation guard, closed generic solution-space shift, and zero five-component target residual
32	source-origin no-go and underdetermination	corrected-KK static-term guard, Appendix-B current test, component-ratio exclusions, and the generic two-parameter ambiguity

3 Construction of the thirteen equations

The correction functions are

$$\{\delta u_I, \beta_I, \delta\varphi_1, \delta\varphi_2, \delta\mathcal{F}, \delta\mathcal{G}, \delta\mathcal{H}, \delta\mathcal{K}, \delta\omega\}. \quad (16)$$

For each radial Euler–Lagrange equation we form

$$E_A^{(2),\text{lin}}[\delta\mathcal{Y}] + E_A^{(4)}[\mathcal{Y}_0] = 0, \quad (17)$$

where A labels the thirteen field equations. The first term is the linearization of the two-derivative equation. The second is the Euler–Lagrange variation of the complete four-derivative invariant evaluated on the uncorrected rotating background. This separation matters computationally: the first term contains all unknown correction functions but remains linear, while the second can be very large but contains no correction functions.

Archival stage. Notebook 01 stores all thirteen $E_A^{(2),\text{lin}}$, all thirteen $E_A^{(4)}$, and the thirteen sums in Eq. (17). Its size is about 597 MiB. It is the package’s explicit equation archive, not the preferred interactive solver. The default workflow in notebook 00 does not rerun the full curvature construction and variation: those stages are explicitly marked `SKIPPED/RESOURCE_BOUND`. They are never counted as successful from-action regeneration.

Structural checks. The rotating two-derivative background gives 13 exact zeros. Removing every correction field from the homogeneous linearization also gives 13 exact zeros. Every equation is linear in correction fields and their derivatives. The notebook-only rational radial reparametrization was checked with exact identities where practical and high-precision independent probes for the largest source expressions.

4 Source extraction and the F^4 question

The four-derivative invariant was evaluated sector by sector. The separate curvature-free BPS vector invariant used as a regression direction has vanishing full first-order Euler–Lagrange variation on the rotating BMPV background. This is the precise special cancellation anticipated for that separate invariant. It does not permit vector terms to be removed from the active curvature-squared invariant: the supersymmetric completion contains nonzero RF^2 , F^4 , scalar, and mixed Chern–Simons pieces.

DISCOVERY: separating sectors localized normalization and sign errors far more efficiently than simplifying the full source at once. PROOF: the final source is the sum of every sector in the active invariant, and the closed solution is substituted into that complete sum.

5 Integrated electric equations and the radial constraint

All three electric equations are exact total derivatives. Integration gives three first integrals

$$\mathfrak{P}_I(r; \delta\mathcal{Y}, \delta\mathcal{Y}') = \text{constant}. \quad (18)$$

Because the action contains Chern–Simons terms, these conserved integrals are the Page representatives of the physical electric charges. We work at fixed physical charges $Q_I = \pi\mu_I/(4G_5)$, so no charge-shifting integration mode is retained. The coefficient matrix of the three electric derivatives is diagonal and nonsingular throughout the physical exterior, so the integrated equations determine $\delta u'_I$. The radial Einstein constraint is algebraic in $\delta\mathcal{G}$ after the areal choice $\delta\mathcal{H} = 0$, so it supplies an algebraic elimination rule. The generic rotation coefficients pass the raw EOM directly in notebook 12. Notebook 17 embeds the original radial equation and all three Page rules, reconstructs the $\delta\mathcal{G}$ rule, and proves its equality with the stored coefficient exactly at generic $\mu_1, \mu_2, \mu_3, r, j$. Its coefficient is linear in j^2 ; positivity at both endpoints of the BMPV interval proves it is nonzero throughout the physical exterior. The same notebook independently substitutes the closed j^2 fields in the unreduced radial equation and obtains literal zero.

The already defined fields β_I remove kinematic mixing with the background fibre and expose rotation parity. The full generic reduced system has eight second-order fields plus three first-order electric fields, hence nineteen first-order modes.

6 The equal-charge benchmark

Before attacking arbitrary charges, the locus $\mu_1 = \mu_2 = \mu_3 = \mu$ was solved exactly. One λ_3 direction still distinguishes species 3, so the correct decomposition contains both singlet and nonsinglet fields. The resulting rational functions terminate at j^3 and solve all thirteen original equations before Page integration and constraint elimination.

This benchmark served three roles. It fixed the physical branch and gauge, provided exact target expressions for every generic charge formula, and first suggested that the rotation hierarchy might terminate. It was a stringent regression test, not a proof that the arbitrary-charge answer had the same finite degree.

7 Closing the generic static seed

The generic $j = 0$ fields first served as regression data for the rotating reconstruction. Notebook 13 makes this sector self-contained by embedding the complete unequal-charge seed and all thirteen static homogeneous/source pairs. Positive-branch rewriting converts the fractional harmonic-function powers to rational functions. For each equation the homogeneous and source members are put over exact denominators, cross-multiplied, and expanded as polynomials in r^2, μ_1, μ_2, μ_3 . All thirteen cleared numerators are literal zero.

The paper uses the harmonic/isotropic-like radius ρ , whereas the solver uses the areal radius r . Notebook 13 contains an embedded proof of Eq. (14). For v4, notebook 28 independently transcribes and checks seven published α' coefficients, then proves both the paper-to-areal and areal-to-paper transformations for every surviving field. It also verifies that the only new common denominator is

$$\sum_{I < J} (\mu_I + r^2)(\mu_J + r^2) > 0 \quad (19)$$

on the physical exterior. Thus the static origin and the reduced EOM certificate meet field by field rather than only through sampled regression. Professor Liu’s coordinate observation is exactly correct, and it does not alter the Lagrangian or any parity-odd convention.

8 Generic unequal charges at order j

After subtracting the exact static seed, the rotation-odd subsystem involves $\{\beta_1^{(1)}, \beta_2^{(1)}, \beta_3^{(1)}, \delta\omega^{(1)}\}$. Its principal differential matrix is nonsingular in the physical exterior. The angular equation also admits a first integral. It was used as an internal consistency relation, while the final four functions were verified directly in the original differential equations.

DISCOVERY: the source was compressed coefficientwise and its rational primitive found before solving the remaining fibre equations. Exact collocation at unequal rational charge triples produced candidate rational functions. The denominators and permutation structure exposed the common combination

$$\Delta(r) = 3r^4 + 2(\mu_1 + \mu_2 + \mu_3)r^2 + \mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3. \quad (20)$$

PROOF: the reconstructed symbolic functions were substituted with μ_1, μ_2, μ_3 arbitrary into all four odd equations, giving four literal zeros. The remaining nine equations vanish at odd order by exact parity. The solution has no $1/r^2$ correction to the angular-momentum tail.

9 Generic order j^2 : compression and reduction

The even fields are

$$\{\delta u_1^{(2)}, \delta u_2^{(2)}, \delta u_3^{(2)}, \delta\varphi_1^{(2)}, \delta\varphi_2^{(2)}, \delta\mathcal{F}^{(2)}, \delta\mathcal{G}^{(2)}, \delta\mathcal{K}^{(2)}\}. \quad (21)$$

Source extraction was the apparent bottleneck. Coefficientwise convolution avoided repeatedly combining giant rational expressions:

equation	original terms	selected j^2 terms	compact monomials
temporal Einstein	25,042	7,671	511
radial Einstein	13,230	3,518	343
angular trace	31,027	10,978	512
squashing	29,782	10,783	512

Every compression passed independent exact rational-point checks. Page and constraint substitution then reduced the nine raw equations to the physical block.

At the representative exact charge point $(\mu_1, \mu_2, \mu_3) = (2, 3, 4)$, the electric equations vanish after Page composition. The principal rank is four in $\{\delta\varphi_1^{(2)}, \delta\varphi_2^{(2)}, \delta\mathcal{F}^{(2)}, \delta\mathcal{K}^{(2)}\}$; the angular-trace equation is a consistency constraint. The squashing equation decouples,

$$(\delta\mathcal{K}^{(2)})'' + \frac{3}{r}(\delta\mathcal{K}^{(2)})' - \frac{8}{r^2}\delta\mathcal{K}^{(2)} = 0, \quad (22)$$

and physical boundary conditions select $\delta\mathcal{K}^{(2)} = 0$. The actual differential core is the three-field rational system $\{\delta\varphi_1^{(2)}, \delta\varphi_2^{(2)}, \delta\mathcal{F}^{(2)}\}$.

10 What failed and what succeeded at order j^2

10.1 Direct symbolic integration

DISCOVERY: a direct coupled `DSolve` request was allowed a bounded ten-minute window. It did not produce a useful closed solution. Intermediate normal form introduced apparent polynomial singularities, and a rational ansatz restricted only to the physical harmonic-function pole structure was inconsistent through tested orders four, five and six. These failures ruled out an overly narrow ansatz; they did not rule out a rational physical solution because apparent factors can cancel only after the coupled system is reconstructed.

10.2 Exact rational collocation

The successful route was:

1. specialize to several exact unequal charge triples;
2. write a sufficiently general rational ansatz for the reduced fields;
3. evaluate the linear equations at enough exact rational values of r^2 to obtain an overdetermined system;
4. solve that system over the rationals;
5. check unused points and then the full symbolic residual;
6. factor numerators and denominators to identify H_I , $\Delta(r)$, and the charge-symmetric polynomials $e_2 = \sum_{I < J} \mu_I \mu_J$ and $e_3 = \mu_1 \mu_2 \mu_3$;
7. reconstruct charge-dependent coefficients from multiple samples;
8. substitute the generic formulas back into the arbitrary-charge equations.

Using too few collocation points gave an underfit one-dimensional nullspace and failed at unused points. Increasing the point count removed the spurious mode. This is an important reproducibility warning: collocation was used for discovery, never as the final proof.

PROOF: the reconstructed fields give literal zero in all eight compact reduced equations and in the independent algebraic $\delta\mathcal{G}^{(2)}$ rule, with symbolic charges. The three electric expressions were then recovered by integrating the Page rules, not guessed independently. All eight equal-charge limits match the prior benchmark.

11 Generic order j^3

The cubic rotation-odd subsystem again contains four fields. Before compression its equation/source leaf-count pairs were

$$(16428, 29247), (16428, 29247), (20352, 36183), (9268, 16201). \quad (23)$$

Exact sample reconstruction and generic charge lifting produced

$$\beta_1^{(3)} = \beta_2^{(3)} = \beta_3^{(3)} = 0, \quad \delta\omega^{(3)} = \frac{24}{r^4(r^2 + \mu_3)\Delta(r)}. \quad (24)$$

PROOF: substitution into the four symbolic cubic equations gives four literal zeros. The equal-charge limit gives $8/[r^4(r^2 + \mu)^3]$, independently reproducing the benchmark.

12 Coefficientwise proof of exact rotation closure

The next question was whether uncomputed fields at $j^4, j^5, \dots$ were required. Setting all such new fields to zero gives

order	equations	result
4	nine even	9 exact zeros
5	four odd	4 exact zeros
6	nine even	9 exact zeros
7	four odd	4 exact zeros
8	nine even	9 exact zeros

Notebook 12 recomputes all 35 residual zeros from separate homogeneous and source members. Canonical member simplification finds five nontrivial j^4 cancellations in $\{\delta u_1, \delta u_2, \delta u_3, \delta \varphi_1, \delta \mathcal{F}\}$ and 30 cases in which both members vanish separately. The raw homogeneous-operator degrees are at most two for even equations and one for odd equations; after inserting the finite solution their contributions reach at most four and three. The full source degrees are

$$\{\delta u_1, \delta u_2, \delta u_3, \delta \varphi_1, \delta \varphi_2, \beta_1, \beta_2, \beta_3, \delta \mathcal{F}, \delta \mathcal{G}, \delta \mathcal{H}, \delta \mathcal{K}, \delta \omega\} = \{4, 4, 4, 4, 2, 3, 3, 3, 6, 6, 8, 8, 7\}. \quad (25)$$

Hence no coefficient above j^8 can be generated at order λ_3 ; the table exhausts the rotation hierarchy.

This is the step that promotes the finite expression from “known through j^3 ” to “exact in j at first order in λ_3 .”

13 Physical normalization

The Page conditions hold the physical electric charges fixed,

$$Q_I = \frac{\pi \mu_I}{4G_5}. \quad (26)$$

There is an exact homogeneous rotation-parameter mode $\delta \omega_{\text{hom}} = c_j/(2r^2)$. It is the infinitesimal parameter shift $j \mapsto j + \lambda_3 c_j$, not a new black-hole hair. The particular solution has no $1/r^2$ correction tail. Notebook 14 then evaluates the complete generic asymptotic Wald–Tachikawa angular surface term. It reconstructs the covariant curvature tensor and the mixed $A^3 \wedge R \wedge R$ tensor with the first Riemann pair lowered, projects both to the algebraic Riemann symmetry class, and stabilizes the asymptotic expansion at four independent orders. Every potentially nondecaying four-derivative coefficient vanishes; the Einstein/Komar calibration gives

$$J_1 = J_2 = \frac{\pi j}{4G_5}. \quad (27)$$

Together with the exact absence of mass, charge, and angular parameter tails, this proves $\delta Q_I = 0$, $\delta J_1 = \delta J_2 = 0$, and $c_j = 0$ at fixed generic physical charges and angular momenta.

14 Horizon regularity

The apparent raw poles were next tested as a geometric question rather than as coordinatewise finiteness of the original functions. Notebook 15 finds one common perturbative horizon shift,

$$\delta(r_H^2) = \frac{16(3j^2 + 2\mu_1\mu_2\mu_3)}{\mu_3(\mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3)}. \quad (28)$$

After centering the radial coordinate on this shifted horizon, it constructs future-horizon ingoing, co-rotating coordinates. For $\mu_I > 0$ and $j^2 < \mu_1\mu_2\mu_3$, every metric generator is analytic, the

horizon block is nondegenerate, and scalar kinetic and gauge-field invariants are finite. Writing χ for the horizon generator and ξ_ψ for the Hopf-fibre Killing vector in the regular co-rotating chart, direct limits show that χ^2 , $\chi \cdot \xi_\psi$, and the radial derivative of χ^2 vanish at orders zero and one, with $\Omega_H^{(1)} = \kappa^{(1)} = 0$. The proof is perturbative at fixed positive gap and is not uniform at the zero-area bound.

15 Independent Cassani/heterotic source check and Eq. (3.31)

Notebook 16 confirms the Lagrangian interpretation stated at the start. The project uses one independent Cassani standard-Weyl STU invariant, λ_3 ; a cyclic sum is not required. The heterotic gravitational normalization selects $\lambda_1 = \lambda_2 = 0, \lambda_3 = \alpha'/8$, while two curvature-free vector invariants remain in the full action. Let $\tilde{F}^I$ denote the rescaled field strengths and e^0 the timelike orthonormal one-form. On rotating BPS,

$$\tilde{F}^I - \tilde{F}^J = -e^0 \wedge d \log(X^I/X^J), \quad (29)$$

and the $d\Theta$ terms cancel without any expansion in j . Symbolic differentiation proves that the extra invariants and their first derivatives with respect to the field strength and scalar gradient vanish. Their metric, vector, and scalar sources are therefore zero, so the heterotic and Cassani first-order sources coincide exactly in rotation.

The same notebook substitutes the arXiv:2606.31987 Eq. (3.31) profiles into the two Bianchi ODEs and the Einstein ODE, obtaining three zeros and no asymptotic charge or rotation tail. It also confirms that the printed constant 1 in W violates the source's own boundary condition; the consistent expression begins with $a/(2r^2)$. The raw two-derivative KK/Weyl metric and scalar dictionary is exact.

16 Direct five-dimensional entropy

Notebook 20 evaluates the corrected horizon determinant after the unique horizon-centering shift. Define

$$P = \mu_1 \mu_2 \mu_3, \quad S_0 = \frac{\pi^2}{2G_5} \sqrt{P - j^2}. \quad (30)$$

The corrected-area contribution is

$$\frac{\delta S_{\text{area}}}{S_0 \lambda_3} = \frac{4(2j^4 - 7j^2 P + 4P^2)}{P \mu_3 (P - j^2)}. \quad (31)$$

The raw BMPV coframe mixes the timelike normal with the Hopf fibre. A constant boost with $u = j/\sqrt{P}$ gives the adapted curvature and electric components. Their covariant Wald kernel collapses to $1/\mu_3$, so

$$\frac{\delta S_{\text{cov}}}{S_0 \lambda_3} = -\frac{4}{\mu_3}. \quad (32)$$

For the mixed term, the essential conventions are $\text{tr}(R \wedge R) = -R^{ab} \wedge R_{ab}$ and $\tilde{\epsilon}^{tr\theta\phi\psi} = +1$. Because this coordinate order reverses the physical orientation, the component interaction $+\frac{\lambda_3}{2e} \tilde{\epsilon} R R A^3$ is $+2\lambda_3 A^3 \wedge \text{tr}(R \wedge R)$ in oriented form language and has the following physical Tachikawa coefficient. Here Γ_N is the pullback of the horizon normal-bundle connection:

$$\delta S_{\text{CS}} = +\frac{\lambda_3}{G_5} \int_{H_{\text{phys}}} \Gamma_N \wedge F^3 = -\frac{\lambda_3}{G_5} \int_{H_{\text{coord}}} \Gamma_N \wedge F^3, \quad \frac{\delta S_{\text{CS}}}{S_0 \lambda_3} = \frac{8j^2}{P \mu_3}. \quad (33)$$

Here $\int_{H_{\text{phys}}} \sigma_L^3 \wedge d\sigma_L^3 = +16\pi^2$, whereas the coordinate-measure value used by the code is $-16\pi^2$. The Tachikawa prefactor and horizon-orientation corrections cancel numerically. No sign of F^3 is reversed. The sum is

$$S = S_0 \left[1 + \frac{4\lambda_3(3P - 4j^2)}{\mu_3(P - j^2)} \right] + \mathcal{O}(\lambda_3^2), \quad (34)$$

which has the asymptotic-charge spin dependence derived in the next section. Notebook 21 does not execute notebook 20: it rebuilds the coordinate Riemann tensor, horizon frame, normal connection, area determinant and trace conversion. A fresh kernel evaluates ten of ten cells with no messages and all six named checks true.

17 Why published rotating entropy formulas disagree

Having fixed the direct result, define

$$\varpi^2 = \frac{J^2}{Q_1 Q_2 Q_3}, \quad 0 \leq \varpi^2 < 1, \quad (35)$$

with Q_I and $J = J_1 = J_2$ held fixed at infinity. Four expressions that are sometimes compared as if they were the same correction are

$$g_{\text{Euler}} = 1 + \frac{\varpi^2}{3}, \quad g_{4\text{D}/5\text{D}} = \frac{1}{1 - \varpi^2}, \quad g_{\text{horizon}} = 1, \quad g_{\text{asym}} = \frac{1 - \frac{4}{3}\varpi^2}{1 - \varpi^2}. \quad (36)$$

They answer different variational questions. An Euler-density Wald term on an uncorrected geometry omits the induced displacement of the solution. Near-horizon/off-shell and 4D/5D formulas naturally use horizon parameters or shifted charges. Write the entropy at fixed solution parameters as $S = S_0 + \lambda_3 S_1 + \mathcal{O}(\lambda_3^2)$, and let ΔQ_I and ΔJ denote the induced first-order shifts of the asymptotic charges and angular momentum. Transforming instead to fixed asymptotic physical charges and angular momentum gives

$$\delta S|_{Q,J} = S_1 - \sum_I \frac{\partial S_0}{\partial Q_I} \Delta Q_I - \frac{\partial S_0}{\partial J} \Delta J. \quad (37)$$

The derivative of the square root enhances these variable redefinitions near the BMPV spin bound, which is why the denominators differ. Notebook 18 proves the algebraic relations and shows that the exact heterotic entropy formula of arXiv:2606.31987 expands to g_{asym} , agreeing with the direct five-dimensional result.

18 Advancing the rotating ten-to-five-dimensional field map

In this section the heterotic and STU parameters are related by

$$(\mu_1, \mu_2, \mu_3) = (q_+, q_-, q_0), \quad j = a, \quad \lambda_3 = \frac{\alpha'}{8}. \quad (38)$$

The corresponding harmonic functions are

$$H_A(r) = 1 + \frac{q_A}{r^2}, \quad A \in \{+, -, 0\}. \quad (39)$$

We write

$$\omega_0(r) = \frac{a}{2r^2}, \quad \boldsymbol{\omega}_0 = \omega_0 \sigma_L^3, \quad \Theta_0 = dt + \boldsymbol{\omega}_0, \quad (40)$$

so the bold symbol is always the rotation one-form; unbold ω below is reserved for a Lorentz connection.

The complete covariant field-redefinition chain is known: ten-dimensional heterotic string-frame fields are first reduced to five-dimensional string-frame variables, corrected by the Bergshoeff–de Roo redefinitions, dualized, Weyl-rescaled to Einstein frame, and finally related to the Riemann-frame STU variables. The obstruction was therefore not the absence of a formal dictionary but its rotating component evaluation. In this section Φ denotes the five-dimensional heterotic generalized dilaton; it is distinct from the STU scalars φ_1, φ_2 . We denote the Kaluza–Klein circle scalar by

Σ , and the five-dimensional momentum and winding potentials by $\mathcal{A}$ and $\mathcal{B}$, their field strengths by $\mathcal{F} = d\mathcal{A}$ and $\mathcal{G} = d\mathcal{B}$, and their combinations by $\mathcal{F}^\pm = e^\Sigma \mathcal{F} \pm e^{-\Sigma} \mathcal{G}$. The symbol h_{STU} denotes the gauge-invariant horizontal three-form in the five-dimensional STU convention; its leading expression is derived below.

The calculation below follows that transformation order:

ten-dimensional string frame $\longrightarrow$ raw five-dimensional quotient $\longrightarrow$ corrected Kaluza–Klein fields
 $\longrightarrow$ torsion-free predual fields $\longrightarrow$ dualized fields $\longrightarrow$ Einstein/Riemann-frame STU fields.

Notebook numbers reflect audit history rather than the order of the field transformations, so the narrative follows the transformations instead.

Notebook 19 performs the exact-in- a rotating KK reduction. With T_{full} denoting the full trace, including its static part, the two trace corrections agree,

$$T_{\text{KK}} = T_{\text{full}} = -\frac{q_+ q_- - 2a^2/(r^2 + q_0)}{(r^2 + q_+)(r^2 + q_-)(r^2 + q_0)}, \quad (41)$$

so their difference cancels from the Kaluza–Klein circle scalar:

$$\Sigma = \frac{1}{2} \log \left(\frac{H_+}{H_-} \right) + \mathcal{O}(\alpha'^2). \quad (42)$$

This is exact in a at first order in α' . The same notebook uses the heterotic base convention $\text{vol}_4 = -(r^3/8) dr \wedge \sigma_L^1 \wedge \sigma_L^2 \wedge \sigma_L^3$, equivalently $\star_4(dr \wedge \sigma_L^3) = -(r/2) \sin \theta d\theta \wedge d\phi$, and derives the self-dual rotating correction. Writing the corrected source profile as $W = \omega_0 + \alpha' w_1 + \mathcal{O}(\alpha'^2)$, it obeys

$$w'_1 + \frac{2w_1}{r} = -\frac{4aq_0}{r(r^2 + q_0)^3}. \quad (43)$$

Notebook 22 completes the winding/Lorentz–Chern–Simons substitution. A source-level recheck first distinguishes the mixed Lorentz matrix used in the CS trace from the two-raised-index connection in the KK shift, and fixes $\tilde{\Omega}_+ = \omega + h_{\text{STU}}/2$. The corrected winding-vector potential is

$$\mathcal{B}_{\text{KK}} = -H_-^{-1} \Theta_0 + \alpha' (b_t^{(1)} dt + b_3^{(1)} \sigma_L^3) + \mathcal{O}(\alpha'^2), \quad (44)$$

with

$$b_t^{(1)} = -\frac{2r^2[-2a^2 + q_- q_+(q_0 + r^2)]}{(r^2 + q_-)^2(r^2 + q_+)(r^2 + q_0)^2}, \quad (45)$$

$$b_3^{(1)} = \frac{a[2a^2 - 2q_- q_+ q_0 - (q_- q_+ + q_- q_0 + q_+ q_0)r^2 - q_0 r^4]}{(r^2 + q_-)^2(r^2 + q_+)(r^2 + q_0)^2}. \quad (46)$$

The winding-vector field strength is closed and $SU(2)$ -invariant, and the displayed potential is an invariant zero-tail representative. This revised formula supersedes the earlier compact multiple of Θ_0 and the intermediate formula that omitted essential a^2 and a^3 terms.

Notebook 24 fixes the other convention-sensitive inputs. In mixed Lorentz indices,

$$\Omega_{+, \text{STU}} = -\omega_{-, \text{het}}, \quad R_{\text{STU}} = -R_{\text{het}}, \quad (47)$$

so curvature squares are unchanged and α' is not sign-flipped. The two Hodge stars that occur during the signature/frame conversion must not be denoted by the same object. With $H_0 = 1 + q_0/r^2$,

$$\mathcal{M}_0^{\text{het}} := \star_{4, \text{het}} dH_0 = -\frac{q_0}{4} \sigma_L^1 \wedge \sigma_L^2 \wedge \sigma_L^3, \quad \mathcal{M}_0^{\text{STU}} := \star_{4, \text{STU}} dH_0 = +\frac{q_0}{4} \sigma_L^1 \wedge \sigma_L^2 \wedge \sigma_L^3 = -\mathcal{M}_0^{\text{het}}. \quad (48)$$

Therefore the leading horizontal field is unambiguously

$$h_{\text{STU}}^{(0)} = -\mathcal{M}_0^{\text{het}} + (H_+ H_-)^{-1} d\omega_0 \wedge \Theta_0 = \mathcal{M}_0^{\text{STU}} + (H_+ H_-)^{-1} d\omega_0 \wedge \Theta_0. \quad (49)$$

With the positive potential of Eq. 4.1, $F^{3(0)} = d(H_0^{-1}\Theta_0)$, the direct five-dimensional Hodge calculation gives

$$e^{2\Phi_0} \star_5 F^{3(0)} = -\mathcal{M}_0^{\text{STU}} - (H_+ H_-)^{-1} d\omega_0 \wedge \Theta_0, \quad h_{\text{STU}}^{(0)} = -e^{2\Phi_0} \star_5 F^{3(0)}. \quad (50)$$

This minus sign is required by the final multiplier; $\mathcal{C}$ is its one-form multiplier and becomes the third STU potential after dualization:

$$+\mathcal{C} \wedge (dh + \mathcal{F} \wedge \mathcal{G}), \quad (F^1, F^2, F^3) = (\mathcal{F}, -\mathcal{G}, d\mathcal{C}), \quad (51)$$

together with $\star_5^2 = -1$ on Lorentzian three-forms. Equivalently, $dh = F^1 \wedge F^2$ and $d(e^{2\Phi} \star_5 F^3) + F^1 \wedge F^2 = 0$. The corrected algebraic relation used below is therefore

$$\tilde{h}_{\alpha\beta\gamma} = -\frac{1}{2}e^{2\Phi_{\text{TF}}} \epsilon_{\mu\nu\alpha\beta\gamma} F^{3\mu\nu} + \frac{\alpha'}{8} \mathcal{J}_{\alpha\beta\gamma}, \quad \mathcal{J}_{\alpha\beta\gamma} := \frac{\delta \mathcal{L}_{4\partial}}{\delta h_{\alpha\beta\gamma}}. \quad (52)$$

Here $\tilde{h}$ is the torsion-free action three-form before it is eliminated. The sign of the $\mathcal{J}$ term is unchanged. At leading order $\tilde{h}^{(0)} = h_{\text{STU}}^{(0)}$. On the static background, the two sides read

$$(e^{2\Phi_0} \star_5 F^{3(0)})_{\theta\phi\psi} = -\frac{q_0}{4} \sin \theta, \quad \tilde{h}_{\theta\phi\psi}^{(0)} = +\frac{q_0}{4} \sin \theta. \quad (53)$$

The source history contains both multiplier conventions, so the final action, the Page equation and this component check are used together rather than combining a displayed line from the earlier convention with the final positive A^3 .

We call the first, derivative-removing map from the corrected Kaluza–Klein variables to the torsionful predual frame Δ_1 . The subsequent map that replaces the torsionful five-dimensional Chern–Simons representative by the Levi–Civita representative is Δ_2 . Both names refer to complete field redefinitions, not to extra dynamical fields.

The predual correction three-form $\mathcal{Q}$ has four nonzero components. Their signs in the package orientation are

$$\mathcal{Q}_{tr\phi} < 0, \quad \mathcal{Q}_{tr\psi} < 0, \quad \mathcal{Q}_{t\theta\phi} < 0, \quad \mathcal{Q}_{\theta\phi\psi} < 0. \quad (54)$$

The same notebook proves that both Δ_1 and Δ_2 are applied with $+\alpha'/8$. It also verifies that the raw quotient metric carries the required inverse H_0 factor in front of its leading five-dimensional block; omitting that factor gives the wrong leading metric.

The next calculation, notebook 23, differentiates the quartic- h , Rh^2 , quadratic-matter, and linear-matter sectors separately before forming their sum. This evaluation order matters. At $(q_-, q_+, q_0, a) = (3, 5, 7, 0)$, the old monolithic output was only $-1715 \sin \theta / (7 + r^2)^3$, the quartic- h member; it omitted the nonzero Rh^2 member. The corrected full result is

$$\mathcal{J}_{\theta\phi\psi}^{\text{full}} = -\frac{98(28 + 3r^2) \sin \theta}{(7 + r^2)^3}. \quad (55)$$

The notebook proves componentwise sector recombination without reading the proposed third vector. Varying the kinetic and four-derivative terms with respect to the same independent anti-symmetric component also proves that the source enters directly, with no extra division by 3!

Notebook 25 independently reconstructs the separate lower-two-form shift. The forward definition is $b_{\text{new}} = b_{\text{old}} - \alpha' \text{Tr}(\omega \wedge h)/8$, but the static source-to-target comparison traverses the inverse relation. Together with inverse-then-direct traversal of the two post-dualization vector changes, this gives the net static shift $\delta \mathcal{C}_\mu = F_{\mu\nu}^3 \partial^\nu \Phi/3$. This certifies the static three-form and third-vector field-strength anchor.

At this stage the working chain appeared to reproduce the complete static map. A separate target-normalization audit then found that the displayed second postdual scalar map left the exact residual

$$\Delta \Phi_{\text{old}} = \frac{[\partial_r \log(H_+ H_-)]^2}{32H_0}. \quad (56)$$

The result was not a missing factor of eight: notebook 11 stores the λ_3 scalar coefficient, and dividing that coefficient by eight after the exact radial pullback gives the published α' coefficient. Reducing the displayed maps to logarithmic derivatives isolated the coefficient inconsistency.

Notebook 26 derives the required term from the leading field strength:

$$e^{-8\Phi_0/3}(\mathcal{F}^{-(0)})^2 = -\frac{2[\partial_r \log(H_+ H_-)]^2}{H_0}. \quad (57)$$

The corrected second postdual map is therefore

$$\Delta_2 \Phi \longrightarrow \Delta_2 \Phi + \frac{1}{8} e^{-8\Phi/3} (\mathcal{F}^-)^2, \quad (58)$$

before the overall $\alpha'/8$. The internal $(\mathcal{F}^-)^2$ coefficient changes from $-3/8$ to $-1/4$; the $(\mathcal{F}^+)^2$ coefficient remains $-3/8$. Equivalently, the added term at the α' -coefficient level is $+e^{-8\Phi/3}(\mathcal{F}^-)^2/64$. The generic static identity is symbolic. The initial rotating reconstruction passed at

$$(3, 5, 7, 1), (2, 3, 5, 1), (4, 7, 9, 2), (1, 6, 8, 3/2), (5, 8, 11, -2),$$

exactly as rational functions of r . These five samples were discovery tests, not the final proof. Notebook 30 subsequently keeps q_-, q_+, q_0, a, r symbolic. With

$$D_- = r^2 + q_-, \quad D_+ = r^2 + q_+, \quad D_0 = r^2 + q_0, \quad \mathcal{A}_q = (q_- + q_+)r^2 + 2q_- q_+, \quad (59)$$

the old mapped-source minus pulled-back target is

$$\mathcal{R}_\Phi = \frac{\mathcal{A}_q^2}{8D_-^2 D_+^2 D_0} - \frac{a^2}{D_- D_+ D_0^2}. \quad (60)$$

Direct contraction of the leading rotating field strength gives

$$\frac{1}{64} e^{-8\Phi_0/3} (\mathcal{F}^{-(0)})^2 = -\mathcal{R}_\Phi. \quad (61)$$

Thus the corrected residual is identically zero for arbitrary unequal charges and rotation. Because the repair is postdual, it is not inserted into the raw or torsion-free predual scalar.

Before finding the correct organization of the third-vector calculation, the inverse- $\hbar$ regression was rerun at the correct stage. The static torsion-free three-form still passes and the rotating one still fails. For $(q_-, q_+, q_0, a) = (3, 5, 7, 1)$, the full-chain closure residual at $r = 2$ is

$$\frac{46775306}{5752902771}. \quad (62)$$

Only triples 125, 134, 345 are nonzero in the inverse-target residual. Thus that particular assembled three-form-first candidate is exactly nonclosed.

Notebook 27 next embeds the exact coefficient matrices for a joint constant-sector solve. If twelve raw occurrences are varied independently, the system has rank $14 = 14$ and a twelve-dimensional fit. This is nonphysical because repeated descendants of each Eq. (3.31) primitive profile must inherit a common coefficient. Tying the four profiles gives ranks 12/13; fixing them to unit normalization gives 11/12. Both are inconsistent. Adding the scalar repair at the wrong predual stage remains inconsistent and is kept only as a negative control.

Notebook 31 resolves the solution-map question by reversing the order of the last calculation. It begins with the raw potential

$$\mathcal{C}_{\text{raw}} = Z_0^{-1} (dt + W \sigma_L^3) \quad (63)$$

for arbitrary radial functions Z_0, Z_+, Z_-, W , rather than trying to define the final field strength through the rejected assembled three-form. Using the physical curved epsilon tensor, all ten independent components of

$$h_{\text{raw}} = -e^{2\Phi_{\text{raw}}} \star_{5, \text{phys}} d\mathcal{C}_{\text{raw}} \quad (64)$$

vanish identically. Reversing the sign fails, and $d^2\mathcal{C}_{\text{raw}} = 0$ provides an independent Bianchi guard. This part of the calculation uses no five-dimensional target.

At first order in α' , the raw Eq. (3.31) potential contributes

$$\mathcal{C}_{\text{raw}}^{(1)} = -\frac{q_0^2 r^2}{D_0^4} dt + \frac{aq_0(q_0 + 2r^2)}{2D_0^4} \sigma_L^3. \quad (65)$$

Comparison with the independently EOM-certified five-dimensional solution fixes the remaining closed solution-space shift

$$\Delta\mathcal{C}_{\text{map}} = T_C(r) dt + S_C(r) \sigma_L^3, \quad (66)$$

$$T_C = \frac{a^2 r^2}{2D_- D_+ D_0^3}, \quad S_C = \frac{a[a^2 - q_0 D_- D_+]}{4D_- D_+ D_0^3}. \quad (67)$$

It vanishes in the static limit, has the required rotation parity and asymptotic falloff, and is closed by construction. More importantly, it is recognized exactly in the primitive source profiles of Eq. (3.31):

$$T_C = \frac{1}{8} u_+ u_0 z_{+, \text{rot}}^{(1)}, \quad S_C = \omega_0 T_C - \frac{1}{4} u_0 w_1, \quad (68)$$

where

$$u_+ = H_+^{-1}, \quad u_0 = H_0^{-1}, \quad z_{+, \text{rot}}^{(1)} = \frac{4a^2}{r^2 D_0^2 D_-}, \quad w_1 = \frac{aq_0}{r^2 D_0^2}, \quad \omega_0 = \frac{a}{2r^2}. \quad (69)$$

Finally, notebook 31 adds the already certified postdual vector redefinition

$$\delta\mathcal{C}_\mu^{\text{post}} = \frac{1}{3} F_{\mu\nu}^{3(0)} \partial^\nu \Phi_0 \quad (70)$$

and the radial pullback generated by $\xi(r)$. The master identity is

$$\mathcal{C}_{\text{raw}}^{(1)} + \Delta\mathcal{C}_{\text{map}} + \delta\mathcal{C}^{\text{post}} = \mathcal{C}_{\text{STU,areal}}^{(1)} - \xi(r) \partial_r \mathcal{C}^{(0)}. \quad (71)$$

Its temporal, fibre, all-five-component, profile, and Bianchi residuals are all zero symbolically for generic unequal charges and arbitrary rotation. This completes the exact target match on the rotating ansatz. The numerical coefficients $1/8$ and $-1/4$ were determined through the certified target comparison and only then recognized in the source profiles. Consequently no unique universal off-shell covariant field redefinition for unrelated backgrounds is claimed. Notebook 27 remains in the package as a chronological record of a rejected route, not as the final physical definition.

The most direct source-side alternative was also tested. The source ledger shows that the B.1–B.3 boundary currents use the plain torsion-free/composite derivative $\mathcal{D}$ and contain no independent h . B.4 is the only displayed current among them with explicit three-form dependence. Its possible dual-vector shift is proportional to $\star_5(\mathcal{F}^+ \wedge \mathcal{F}^+)$. On the generic rotating source ansatz $\mathcal{F}^+ = dr \wedge v$, so all five components of this candidate vanish identically. These displayed currents therefore do not reproduce T_C, S_C . The printed corrected KK scalar is also insufficient: using its full static plus rotating expression in a universal vector rescaling would create a forbidden static term. Neither the radial pullback nor the displayed postdual vector redefinition has the required component ratio. Moreover, closure, parity, the static anchor and falloff admit an arbitrary two-parameter family $c_a u_0 \tau_\omega \Theta_0 + c_b u_0 w_1 \sigma_L^3$; they do not select $c_a = 1/4, c_b = -1/4$. An undisplayed equation-of-motion field redefinition or a larger covariant basis would therefore be needed for a universal off-shell derivation.

19 Human reproduction protocol

Read and evaluate the notebooks in the staged verification order below. This is deliberately distinct from the report's chronological physics derivation: notebook 13 is an independent, after-the-fact certificate of the generic static result introduced before rotation in the report. Start a fresh Wolfram kernel for each decisive verifier. A from-EOM reconstruction uses the following sequence:

1. evaluate notebook 00A for the geometric orientation convention and notebook 29 for the retrospective sealed-v1 parity-odd compatibility verdict;
2. inspect notebook 00's ansatz, invariant tables, and stage ledger; note that full curvature/action variation is explicitly resource-bounded;
3. use notebook 01 as the literal equation archive; its complete 69-cell fresh-kernel run passes but requires substantial memory;
4. evaluate notebook 02's integrated electric equations and radial Einstein-constraint archive, then notebook 03's source-sector checks;
5. verify the equal-charge benchmark with notebooks 04–05 and inspect the generic linear-rotation result and normalization in notebooks 06–07;
6. inspect the quadratic and cubic coefficients in notebooks 08–09 and the closure and combined fields in notebooks 10–11;
7. evaluate notebook 12 to substitute those fields into literal j , j^2 , and j^3 equations and into all j^4 – j^8 pairs;
8. evaluate notebook 13 for the generic static endpoint, then notebook 28 for the published paper-radius coefficients, exact two-way coordinate map, and general exact-in- j pull-back; continue with notebooks 14–17 for physical normalization, horizon regularity, the Cassani/heterotic source relation, and the Page-reduced radial identity;
9. evaluate notebooks 20 and 21 for the direct entropy and its independent reconstruction, then notebook 18 for the fixed-variable comparison;
10. evaluate notebook 19 for the raw rotating reduction and scalar-ratio checks, followed by notebooks 22–25 for the corrected winding sector, sector-separated dualization source, convention/static anchor, and independent lower-two-form direction;
11. evaluate notebook 26 for the corrected second postdual scalar map and its generic-static/five-rotating-sample discovery ledger;
12. evaluate notebook 27 for the exact residual and tied-profile rank exclusions of the rejected three-form-first route;
13. evaluate notebook 30 for the arbitrary-charge symbolic rotating scalar proof, then notebook 31 for the target-independent raw duality identity and the exact generic solution-level third-vector target identity;
14. evaluate notebook 32 for the B.1–B.4 source ledger, the B.4 and full-KK-rescaling no-go tests, the direction exclusions, and the two-parameter ambiguity;
15. compare the machine-readable outcomes with the concise ledgers in notebooks 08–11 and the expected outputs printed beside every new audit.

Notebook 12 must return **True**; its log records the low-order passes, all source degrees, 35/35 higher-order residual zeros, and the five nontrivial cancellations. The concise notebooks provide readable formulas, while notebook 12 independently recomputes the decisive rotation cancellations. Historical conditional text in notebook 19 is superseded sector by sector by notebooks 22–32. Notebook 27's nonclosure remains a valid negative diagnostic for its assembled three-form-first object; notebook 31 supplies the physical raw-potential-first completion.

20 Scope and the solution-level field map

Writing $\mathcal{Y}$ for any field in the ansatz, the evidence-aware result is

$$\mathcal{Y} = \mathcal{Y}_{\text{BMPV}}(j) + \lambda_3 \delta \mathcal{Y}_{\text{generic}}^{\text{exact in } j} + \mathcal{O}(\lambda_3^2). \quad (72)$$

The static coefficient, rotation hierarchy, Page/radial identity, generic asymptotic Wald–Tachikawa normalization, and invariant horizon regularity all have standalone exact certificates. The equality $\lambda_3 = \alpha'/8$ is used as the heterotic gravitational normalization; $\mathcal{O}(\alpha'^2)$ is not controlled. The static paper-coordinate solution and its exact two-way map to the areal proof gauge are separately certified. The same pullback gives a paper-matched exact-in- j presentation, without asserting uniqueness of a rotating isotropic gauge. The sealed-v1 parity audit finds no sign reversal. The extra heterotic vector-invariant sources vanish exactly on the rotating BPS background. The corrected map between the five-dimensional Riemann/Einstein variables and the rotating ten-dimensional heterotic ansatz has an exact target match at the solution level. The covariant redefinition chain, raw KK reduction, scalar-ratio cancellation, self-dual rotating correction, leading horizontal ledger, winding-vector/Lorentz–Chern–Simons substitution, the full predual correction three-form $\mathcal{Q}$, pre-dual Δ_1/Δ_2 direction, corrected postdual scalar map, exact static anchors, lower-two-form shift, and sector-separated algebraic corrected-dualization source are proved. Notebook 30 proves the rotating scalar map for arbitrary charges. Notebook 31 proves the raw duality identity for arbitrary radial profiles and the final generic third-vector potential match after a closed solution-space shift, postdual redefinition, and radial pullback. Notebook 32 independently rules out using the printed full corrected-KK scalar as a universal vector rescaling and rules out the displayed Appendix-B boundary currents as source-side derivations of that shift. The literal older three-form-first chain is nonclosed and the tested tied constant-sector repair is excluded, so it is retained only as a rejected route. The remaining qualification is conceptual rather than a missing solution component: the two coefficients in the closed shift have not been derived from a unique universal off-shell covariant field-redefinition formula valid on unrelated backgrounds. The entropy-convention algebra and the direct five-dimensional horizon entropy evaluation are both exact at first order in λ_3 for the regular branch $j^2 < P$; the perturbative formula is not extended to the classical zero-area bound.

21 Late crosswalk for internal notebook symbols

The derivation above uses physical notation. The audited Input cells of the packaged BMPV notebooks retain shorter internal names. They are translated here, after the complete human-readable calculation, so they do not interrupt the reading order:

notebook symbol	report quantity	meaning
x	r^2	notebook-only rational radial variable
q1,q2,q3	μ_1, μ_2, μ_3	harmonic parameters
jrot	j	rotation parameter
U1,U2,U3	$\delta u_1, \delta u_2, \delta u_3$	electric corrections
B1,B2,B3	$\beta_1, \beta_2, \beta_3$	Hopf-fibre gauge corrections
Om	$\delta \omega$	rotation one-form correction
Ft	$\delta \mathcal{F}$	temporal warp correction
Gr	$\delta \mathcal{G}$	radial warp correction
Sq	$\delta \mathcal{K}$	Hopf-fibre squashing correction
P or p	$-\sqrt{6} \delta \varphi_1/4$	internal first scalar coordinate
S or s	$\delta \varphi_2/\sqrt{2}$	internal second scalar coordinate

Thus $\delta \varphi_1 = -4 \delta p/\sqrt{6}$ and $\delta \varphi_2 = \sqrt{2} \delta s$. The abbreviation Sq means *squashing*; it is not a field strength. Notebook filenames retain some compact internal terminology, while the report consistently uses parenthesized rotation orders $\delta \mathcal{Y}^{(n)}$.